

$$\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t} \quad \text{Equation of continuity}$$

$$\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$$

$$+ \frac{\partial}{\partial t} (\epsilon_0 \vec{\nabla} \cdot \vec{E})$$

$$\Rightarrow \vec{\nabla} \cdot \vec{J} + \frac{\partial}{\partial t} \epsilon_0 (\vec{\nabla} \cdot \vec{E}) = 0$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

1888 Hertz experimentally verified.

A changing electric field will introduce a magnetic field.

For loop A2

$$E = \frac{\sigma}{\epsilon_0} = \frac{1}{\epsilon_0} \frac{Q}{A}$$

$$\frac{\partial E}{\partial t} = \frac{1}{\epsilon_0 A} I$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc} + \mu_0 \epsilon_0 \int \frac{\partial \vec{E}}{\partial t} \cdot d\vec{a}$$

$$= \frac{1}{\epsilon_0 A} I \cdot A = \frac{I}{\epsilon_0} \mu_0 \epsilon_0 = \mu_0 I$$

prob 7.31 - 7.35

$$\vec{J}_d = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Displacement current

Maxwell's equations,

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \text{— Maxwell's 4th}$$

$$\vec{F} = q (\vec{E} + \vec{v} \times \vec{B})$$

Classical electrodynamics.

Ex. 7.14

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \text{integrating}$$

$$\Rightarrow B \cdot 4\pi r^2 = 0 \quad \int (\vec{\nabla} \cdot \vec{B}) \rho \bar{v} = 0$$

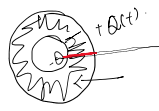
$$\Rightarrow \oint \vec{B} \cdot d\vec{a} = 0$$

$$\left(\frac{\partial}{\partial r} + \frac{\partial}{\partial \theta} + \frac{\partial}{\partial \phi} \right)$$

$$\Rightarrow B = 0 \quad \text{from (a)}$$

$$\vec{J} = \sigma \vec{E} = \sigma \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r}$$

$$\text{current } I = -\dot{Q} = \int \vec{J} \cdot d\vec{a} = \frac{\sigma Q}{\epsilon_0}$$



Material following ohm's law

$$\vec{J}_d = \epsilon_0 \frac{\partial \vec{E}}{\partial t} = \frac{1}{4\pi} \frac{\dot{Q}}{r^2} \hat{r}$$

$$= - \frac{\nabla Q}{4\pi\epsilon_0 r^2} \hat{r}, \text{ displacement current}$$

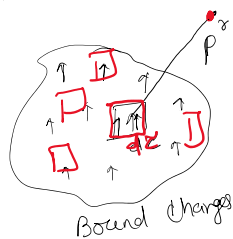
$$\dot{Q} = \frac{dQ}{dt}$$

Dipole materials

$\vec{P} \equiv$ Dipole moment per unit volume

Total number of dipoles in a unit volume $d\tau$, $\vec{P} d\tau$.

$$V(r) = \frac{1}{4\pi\epsilon_0} \int \frac{\vec{P}(\vec{r}') \cdot \hat{r}}{r^2} d\tau'$$



identity, $\nabla' \left(\frac{1}{r} \right) = \frac{\hat{r}}{r^2} = \frac{1}{4\pi\epsilon_0} \int \vec{P} \cdot \nabla' \left(\frac{1}{r} \right) d\tau'$

Integration by parts,

$$\int \vec{P} \cdot \nabla' \left(\frac{1}{r} \right) d\tau' = \int \frac{1}{r} (\nabla' \cdot \vec{P}) d\tau' - \int \frac{1}{r} (\vec{P} \cdot \hat{n}) da'$$

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{1}{r} (\nabla' \cdot \vec{P}) d\tau' + \frac{1}{4\pi\epsilon_0} \int \frac{1}{r} \vec{P} \cdot d\vec{a}'$$

$$V = \frac{Q}{r} = \frac{\int \rho d\tau}{r} = \frac{\int \rho d\tau}{r}$$

Bounded volume charge density

$$\boxed{\begin{aligned} \rho_b &= -\nabla' \cdot \vec{P} \\ \sigma_b &= \vec{P} \cdot \hat{n} \end{aligned}}$$

$\hat{n}$ dirⁿ of surface

Total charge density in the presence of material,

$$\rho = \rho_f + \rho_b$$

bound charge density

magnetic dipoles, $\vec{m}$ - for a single magnetic dipole

$\vec{M} \equiv$ Total magnetic dipole moment per unit volume.

(Magnetization)

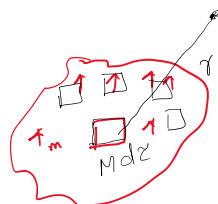
$$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{\vec{M} \times \hat{r}}{r^2} d\tau$$

$$\vec{J} d\tau = \vec{M} d\tau$$

$$\nabla' \left(\frac{1}{r} \right) = \frac{\hat{r}}{r^2}$$

$$= \frac{\mu_0}{4\pi} \int \vec{M} \times (\nabla' \left(\frac{1}{r} \right)) d\tau$$

$$= \frac{\mu_0}{4\pi} \left[\int \frac{1}{r} (\nabla' \times \vec{M}) d\tau' - \int \left(\nabla' \times \left(\frac{\vec{M}}{r} \right) \right) d\tau' \right]$$



$$= \frac{\mu_0}{4\pi} \int \frac{1}{r} (\bar{\nabla}' \times \mathbf{M}) d\tau' - \int \left[\bar{\nabla}' \times \left(\frac{\mathbf{M}}{r} \right) \right] d\tau'$$

as, $\nabla \times \left(\frac{\mathbf{A}}{B} \right) = \frac{1}{B} \nabla \times (\mathbf{A}) + \mathbf{A} \nabla \frac{1}{B}$

$$\begin{aligned} \mathbf{A} &= \mathbf{M} \\ B &= r \end{aligned}$$

$$\mathbf{B} = \frac{\mu_0}{4\pi} \int \frac{1}{r} (\bar{\nabla}' \times \mathbf{M}) d\tau' - \int \frac{1}{r} [\mathbf{M}(r') \times d\mathbf{a}']$$

$$\mathbf{B} = \int \frac{(\bar{\mathbf{J}} \times \hat{\mathbf{r}})}{r} d\tau' \quad \int \frac{\mathbf{I} \times \hat{\mathbf{r}} d\mathbf{l}}{r^2}$$

$$\boxed{\bar{\mathbf{J}}_b \equiv \bar{\nabla} \times \bar{\mathbf{M}}}$$

Total current density, $\bar{\mathbf{J}} = \bar{\mathbf{J}}_f + \bar{\mathbf{J}}_b + \frac{\partial \mathbf{P}}{\partial t}$ - dipole moment per unit volume

$$\bar{\mathbf{J}} = \bar{\mathbf{J}}_f + \bar{\mathbf{J}}_b$$

Gauss's law,

$$\bar{\nabla} \cdot \bar{\mathbf{E}} = \frac{\bar{\rho}}{\epsilon_0}$$

$$= \frac{1}{\epsilon_0} (\bar{\rho}_f - \bar{\nabla} \cdot \bar{\mathbf{P}})$$

$$\Rightarrow (\bar{\nabla} \cdot \bar{\mathbf{E}} + (\bar{\nabla} \cdot \bar{\mathbf{P}})/\epsilon_0) = \frac{\bar{\rho}_f}{\epsilon_0}$$

$$\Rightarrow (\bar{\nabla} \cdot \bar{\mathbf{E}} \epsilon_0 + \bar{\nabla} \cdot \bar{\mathbf{P}}) = \bar{\rho}_f$$

$$\Rightarrow \boxed{\bar{\nabla} \cdot \bar{\mathbf{D}} = \bar{\rho}_f}$$

$$\boxed{\bar{\mathbf{D}} \equiv \epsilon_0 \bar{\mathbf{E}} + \bar{\mathbf{P}}} \text{ - Electric displacement}$$

Ampere's law,

$$\bar{\nabla} \times \bar{\mathbf{B}} = \mu_0 \bar{\mathbf{J}} + \frac{\partial \mathbf{E}}{\partial t} (\epsilon_0 \mu_0)$$

$$= \mu_0 \left(\bar{\mathbf{J}}_f + \bar{\mathbf{J}}_b + \frac{\partial \mathbf{P}}{\partial t} \right) + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

$$= \mu_0 \left(\bar{\mathbf{J}}_f + \bar{\nabla} \times \bar{\mathbf{M}} + \frac{\partial \mathbf{P}}{\partial t} \right) + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

$$\bar{\mathbf{H}} = \frac{1}{\mu_0} \bar{\mathbf{B}} - \bar{\mathbf{M}} \quad \Rightarrow \quad \bar{\nabla} \cdot \bar{\mathbf{H}} = \frac{1}{\mu_0} \bar{\nabla} \cdot \bar{\mathbf{B}} - \bar{\nabla} \cdot \bar{\mathbf{M}} \quad \text{may not be zero.}$$

$$\bar{\nabla} \times \bar{\mathbf{B}} - \mu_0 (\bar{\nabla} \times \bar{\mathbf{M}}) = \mu_0 \bar{\mathbf{J}}_f + \mu_0 \frac{\partial \mathbf{P}}{\partial t} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

$$\Rightarrow \bar{\nabla} \times \left(\frac{\bar{\mathbf{B}}}{\mu_0} - \bar{\mathbf{M}} \right) = \bar{\mathbf{J}}_f + \frac{\partial}{\partial t} \bar{\mathbf{D}}$$

$$\Rightarrow \boxed{\bar{\nabla} \times \bar{\mathbf{H}} = \bar{\mathbf{J}}_f + \frac{\partial}{\partial t} \bar{\mathbf{D}}}$$

$$\bar{\nabla} \cdot \bar{\mathbf{D}} = \bar{\rho}_f$$

$$\bar{\nabla} \times \bar{\mathbf{E}} = - \frac{\partial \bar{\mathbf{B}}}{\partial t}$$

$$\bar{\nabla} \times \bar{\mathbf{B}} \cong \bar{\nabla} \times \bar{\mathbf{H}} = \bar{\mathbf{J}}_f + \frac{\partial \bar{\mathbf{D}}}{\partial t}$$

$$\bar{\nabla} \cdot \bar{\mathbf{B}} = 0$$

Maxwell's differential eqns for conductive matter.

Linear medium

$$\bar{P} = \epsilon_0 \chi_e \bar{E} \quad \& \quad M = \chi_m H$$

$$\bar{D} = \epsilon \bar{E} \quad \text{and} \quad H = \frac{1}{\mu} \bar{B}$$

$$\boxed{\epsilon = \epsilon_0 (1 + \chi_e) \quad \& \quad \mu = \mu_0 (1 + \chi_m)}$$

$$\begin{aligned} \bar{H} &= \frac{1}{\mu_0} \bar{B} - \bar{M} = \frac{1}{\mu_0} \bar{B} - \chi_m \bar{H} \\ \Rightarrow \bar{H} (1 + \chi_m) &= \frac{1}{\mu_0} \bar{B} \Rightarrow \bar{H} = \frac{\bar{B}}{\underbrace{\mu_0 (1 + \chi_m)}_{\mu}} \end{aligned}$$

Boundary conditions for classical ED

Integral form of the maxwell's eqn's

$$\int_V \nabla \cdot \bar{D} dv = \int_V \rho_f dv$$

$$\Rightarrow \oint_S \bar{D} \cdot d\bar{a} = Q_{f \text{ enc}} \quad \text{charge enclosed by the surface}$$

$$\oint_S \bar{B} \cdot d\bar{a} = 0$$

$$\nabla \times \bar{E} = - \frac{\partial \bar{B}}{\partial t}$$

$$\oint (\nabla \times \bar{E}) \cdot d\bar{a} = - \frac{d}{dt} \int \bar{B} \cdot d\bar{a}$$

$$\Rightarrow \oint \bar{E} \cdot d\bar{u} = - \frac{d}{dt} \int \bar{B} \cdot d\bar{a}$$

$$\nabla \times \bar{H} = \bar{J}_f + \frac{\partial \bar{D}}{\partial t}$$

$$\oint (\nabla \times \bar{H}) \cdot d\bar{a} = \int \bar{J}_f \cdot d\bar{a} + \int \frac{\partial \bar{D}}{\partial t} \cdot d\bar{a}$$

$$\Rightarrow \oint \bar{H} \cdot d\bar{u} = I_{f \text{ enc}} + \frac{d}{dt} \int \bar{D} \cdot d\bar{a}$$

free current enclosed by the surface a.

Integral form of Maxwell's eqn's

$$\left. \begin{aligned} \oint_S \bar{D} \cdot d\bar{a} &= Q_{f \text{ enc}} \\ \oint_S \bar{B} \cdot d\bar{a} &= 0 \end{aligned} \right\} \text{over a closed surface}$$

$$\left. \begin{aligned} \oint_{\text{over surface}} \bar{E} \cdot d\bar{u} &= - \frac{d}{dt} \int \bar{B} \cdot d\bar{a} \\ \oint_{\text{over surface}} \bar{H} \cdot d\bar{u} &= I_{f \text{ enc}} + \frac{d}{dt} \int \bar{D} \cdot d\bar{a} \end{aligned} \right\}$$

over surface
bounded
by the
loop.

$$\left\{ \begin{array}{l} \oint_L \vec{E} \cdot d\vec{l} = - \frac{d}{dt} \int_S \vec{D} \cdot d\vec{a} \\ \oint_L \vec{H} \cdot d\vec{l} = I_{f \text{ enc}} + \frac{d}{dt} \int_S \vec{D} \cdot d\vec{a} \end{array} \right.$$

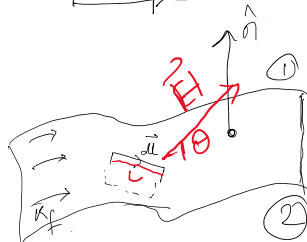
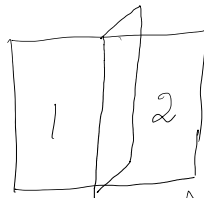
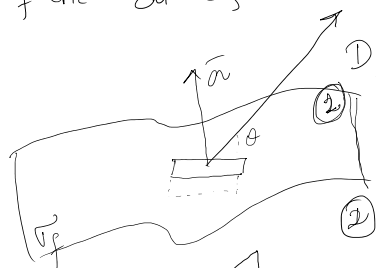
$$\oint \vec{D} \cdot d\vec{a} = \int \sigma_f \cdot d\vec{a}$$

$$\Rightarrow (\vec{D}_1^\perp - \vec{D}_2^\perp) \cdot \hat{n} = \sigma_f$$

$$\Rightarrow \vec{D}_1^\perp - \vec{D}_2^\perp = \sigma_f \hat{n}$$

$$\vec{B}_1^\perp - \vec{B}_2^\perp = 0$$

$$\oint \vec{E} \cdot d\vec{l} = - \frac{d}{dt} \int \vec{B} \cdot d\vec{a}$$



Rough

$$\int da = \int dx dy$$

$$= \int dl \hat{n}$$

$$= L \hat{n}$$

Considering an infinitesimal surface width in the loop,
the magnetic flux is zero.

$$\Rightarrow \oint \vec{E} \cdot d\vec{l} = 0$$

$$\Rightarrow \vec{E}_1'' - \vec{E}_2'' = 0$$

$$\oint \vec{H} \cdot d\vec{l} = \int \vec{J} \cdot d\vec{a} + \frac{d}{dt} \int \vec{D} \cdot d\vec{a}$$

$$\Rightarrow \vec{H}_1 \cdot \vec{l} - \vec{H}_2 \cdot \vec{l} = I_{f \text{ enc}}$$

$$I_{f \text{ enc}} = \vec{K}_f \cdot (\hat{n} \times \vec{l}) = (\vec{K}_f \times \hat{n}) \cdot \vec{l}$$

$$\Rightarrow \vec{H}_1'' - \vec{H}_2'' = (\vec{K}_f \times \hat{n})$$

Hence $\parallel$ component of mag. field is discontinuous.

$$\epsilon_1 E_1^\perp - \epsilon_2 E_2^\perp = \sigma_f \quad \text{--- (a)}$$

$$\vec{D}_1 = \epsilon \vec{E}_1 + \vec{P}$$

$$E_1'' - E_2'' = 0$$

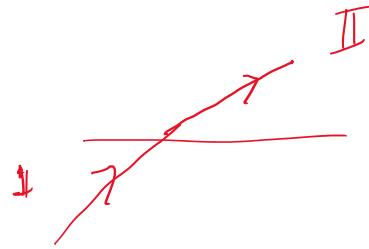
$$n^\perp \quad R^\perp = 0$$

$\perp$ $\perp$

$$B_1^\perp - B_2^\perp = 0$$

$$\frac{1}{\mu_1} B_1'' - \frac{1}{\mu_2} B_2'' = \vec{k}_f \times \hat{n}$$

$$E_1^\perp - E_2^\perp \left(\frac{\epsilon_2}{\epsilon_1} \right) = \frac{\sigma_f}{\epsilon_1}$$

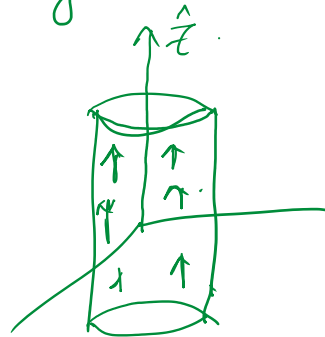


Ex Consider a uniformly magnetized infinite circular cylinder, of radius R , with its axis coinciding with z -axis. The $\vec{M} = M_0 \hat{z}$ inside the cylinder.

(a) Find $\vec{H}(\vec{r})$

(b) Find $\vec{B}(\vec{r})$

$$\vec{M} = M_0 \hat{z}$$



Ampere's law.

$$\vec{\nabla} \times \vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t}$$

$$\vec{\nabla} \times \vec{H} = \vec{J}_f = 0 \quad \text{--- (1)}$$

$$\vec{\nabla} \cdot \vec{B} = \vec{\nabla} \cdot \vec{H} = \vec{\nabla} \cdot \left(\frac{\vec{B}(\vec{r})}{\mu_0} - \vec{M}(\vec{r}) \right) = 0$$

$$\Rightarrow \frac{1}{\mu_0} \vec{\nabla} \cdot \vec{B} - \vec{\nabla} \cdot \vec{M} = 0$$

$$\vec{\nabla} \cdot \vec{H} = 0 \quad \text{--- (2)}$$

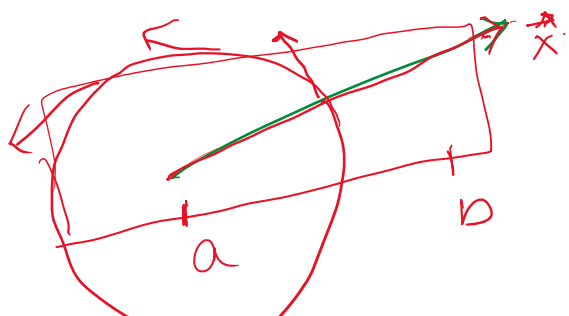
$$\Rightarrow \vec{H} = 0$$

Rough

$$\vec{X} = \vec{X}_{\text{div}} + \vec{X}_{\text{curl}}$$

0 + 0

- - M



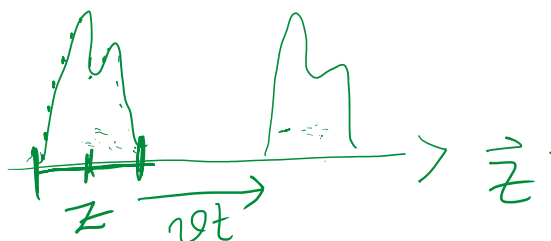
$$\bar{x} = 0$$


$$\vec{H} = \frac{\vec{B}(\vec{r})}{\mu_0} - M(\vec{r}) = 0.$$

$$\Rightarrow \vec{B}(\vec{r}) = \begin{cases} \mu_0 M_0 \hat{z} & r < R \\ 0 & r > R \end{cases}$$

Electromagnetic waves

$$f(z, t) = g(z - vt)$$



$$f(z, t) = g_1(z - vt) + g_2(z + vt)$$

$$\frac{df(z, t)}{dz} = g_1'(z - vt) + g_2'(z + vt)$$

$$\frac{d^2 f(z, t)}{dz^2} = g_1''(z - vt) + g_2''(z + vt) \quad (a)$$

$$\frac{df(z, t)}{dt} = -vg_1'(z - vt) + v g_2'(z + vt)$$

$$\frac{d^2 f(z, t)}{dt^2} = v^2 g_1''(z - vt) + v^2 g_2''(z + vt) \quad (b)$$

$$\boxed{\frac{d^2 f(z, t)}{dz^2} - \frac{1}{v^2} \frac{d^2 f(z, t)}{dt^2} = 0}$$

Any vector with three components $\vec{x}, \vec{y}, \vec{z}$.

$$\nabla^2 f(r, t) = \frac{d^2 f(r, t)}{dx^2} + \frac{d^2 f(r, t)}{dy^2} + \frac{d^2 f(r, t)}{dz^2} - \frac{1}{v^2} \frac{d^2 f}{dt^2}$$

$$\frac{d^2 f(x,t)}{dx^2} + \frac{d^2 f(x,t)}{dy^2} + \frac{d^2 f(x,t)}{dz^2} - \frac{1}{c^2} \frac{d^2 f}{dt^2} = 0$$

$$\Rightarrow \boxed{\nabla^2 f(x,t) - \frac{1}{c^2} \frac{d^2 f}{dt^2} = 0}$$

$$1. \quad \vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t} \quad (1)$$

curl ~~of~~,

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = - \frac{\partial}{\partial t} (\vec{\nabla} \times \vec{B})$$

$$\Rightarrow \vec{\nabla} (\vec{\nabla} \cdot \vec{E}) - \nabla^2 \vec{E} = - \frac{\partial}{\partial t} \left(\mu_0 \vec{J} + \frac{\partial \vec{E}}{\partial t} \mu_0 \epsilon_0 \right)$$

$$\Rightarrow \vec{\nabla} \frac{\rho}{\epsilon_0} - \nabla^2 \vec{E} = - \mu_0 \frac{\partial \vec{J}}{\partial t} - \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$